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Famous_Raj_Bhat_Proposal

Course 608

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Research Topic: Testing How Much Solar Power Forecasts Really Beat Simple Baselines Under Fair Time-Series Validation

Title: Do Solar Forecasting Models Really Beat Simple Baselines? A Leakage-Free Test on European Zone-Level Data

Research Problem:

“Solar power is growing rapidly, but its output varies with weather. Solar plants can lose half their output in minutes when a bank of clouds rolls over. Grid operators balance supply and demand continuously, so they use forecasts to decide how much backup power to keep on hand. A gloomy forecast leads to wasted money on unnecessary backup. If the grid is too rosy, it's at risk.

Many papers claim their solar forecasting model is accurate, but the method to verify such a claim is often weak. The field has reviews pointing out that studies use different baselines and error measures, which makes their results difficult to compare. Three problems keep coming up. First, the baseline is poor or nonexistent. Over the next few hours, persistence (guessing that the next hour will look like this hour) is surprisingly hard to beat, yet it is often skipped or swapped for an easier one. Second, some studies randomly split their data, which means a model learns from hours adjacent to the ones it is tested on, so the score looks better than it would be in actual use. Third, average error disguises abrupt drops in output. These drops are infrequent, but they are precisely what operators dread.

Thus, a low error count on a paper does not tell us whether a model is actually useful. This proposal tests how much of the reported skill holds up when the baselines are strong, the data split respects time, and sudden drops are checked on their own.

Existing Literature

Antonanzas et al. [1] reviewed the PV forecasting methods and found that studies rarely use the same metrics or baselines. Ahmed et al. [2] reached a similar point of view and showed that the weather inputs and preprocessing vary accuracy a lot. Yang et al. [3], a large group of solar forecasting researchers, suggested a standard way to verify forecasts and proposed the RMSE skill score against a reference combining climatology and persistence. That paper lays down the rules but doesn't run a controlled test of modern models under those rules. Hong et al. [4] contributed a shared benchmark (GEFCom2014 competition), but it was a fixed, offline dataset.

On the modeling side, LightGBM [5] and Prophet [6] are popular as they are fast and easy to use. On the validation side, Cerqueira et al. [7] compared approaches to estimate forecast performance and found that out-of-sample methods respecting the time order provided the most accurate estimates when a series evolves over time.

Recent work shows that the problem has not gone away. Mansour et al. [8] tested tree-based models on one Australian solar site and reported skill scores of 97 to 98 percent, but against a persistence baseline built from a fixed lag of 288 steps, which the authors describe as deliberately weak. Kleinebrahm et al. [9] state that accuracy gains in energy forecasting are often not comparable, as studies differ in data, time period, information available, and scoring. To tackle this, they created a live benchmark.

So the gap is this: existing work reviews the field, sets standards, or tunes one family of models. I could not find a single controlled study that combines strong baselines, skill by forecast horizon, size of the random-split effect, and sudden-drop detection on the same data. That’s what this proposal is about.

Research Question

Main question: To check, how much better are our solar PV forecasting models when tested fairly compared to strong baselines, and how much of the often-reported improvement is due to weak baselines and leaky validation?

To answer this question, I will go through three smaller questions:

Q1. For what forecast horizon (1-48 hours) LightGBM and Prophet stop clearly outperforming smart persistence?

Q2. What is the difference in accuracy between a random train/test split and a time-respecting split on the same data/models?

Q3. How well do the models catch sudden drops in output, and how many false alarms does that come with?

My hypotheses are (H1) the models clearly outperform strong baselines at 6 to 48 hours but only slightly at 1 hour; (H2) errors are considerably smaller when using random splitting than a time-respecting split; and (H3) sudden-drop detection is less sensitive than the average error suggests.

Methods

This is a quantative, computational study. It is divided into five parts.

Data. I will use hourly solar generation for two bidding zones, one sunny and one cloudy (e.g., Spain and Germany), from the ENTSO-E Transparency Platform [10] for a period of about five to six years. Weather inputs (solar irradiance, temperature, cloud cover, and wind) are taken from ERA5 reanalysis [11]. This is divided by installed capacity to be able to compare zones. I will remove duplicate timestamps around clock changes, put everything on a single UTC hourly grid, fill in gaps shorter than 3 hours by interpolation, and drop and log gaps longer than that. The training and scoring do not include night hours. If they stay, a model gets thousands of easy correct zeros and looks better than it is. Similarly, I won't use MAPE either; it breaks down near zero output.

Features. Weather, clear-sky index, sun position, calendar terms, past generation (lags of 1, 2, 3, 24, and 168 hours and rolling averages) One rule is strict: a feature at time t can only use information available at time t. I'll create unit tests to verify this. Historical weather is more accurate than a real-time weather forecast, so I will also rerun the day-ahead tests with noise added to the weather inputs.

Baselines and models. The baselines are persistence, smart persistence (fix the clear sky index), seasonal naive (same hour yesterday), climatology, and the combination of climatology and persistence as recommended by Yang et al. [3]. The models are LightGBM (the main model, one for each horizon group), Prophet with irradiance and cloud cover as extra inputs, and linear regression as a simple reference. Horizons are 1, 3, 6, 24, and 48 hours.

Validation. The newest 20% of data is the test set, and it is used only once. Training is performed on the oldest 80% of data, tuning with rolling-origin folds. I leave a gap between training and validation equal to the longest horizon so that lag features cannot

leak. Then, to answer Q2, I repeat the same experiment with a random 80/20 split and compare the two results.

Evaluation. The main measures are RMSE, MAE, capacity-normalized RMSE, and the RMSE skill score, 1 - RMSE_model / RMSE_baseline, reported per horizon against each baseline [3]. I will check that a difference is not just noise with the Diebold-Mariano test [12] and repeat runs with different random seeds. For sudden drops I define an event as a fall of more than 20% of capacity in one hour and report precision, recall, F1, and false alarm rate with a small tolerance on timing. Results will also be presented by season and for clear and cloudy days.

Ethics and limits. All data are public and aggregated, so there are no issues with personal data or approval. There are two limits worth stating now. At the zone level, the data smooths out the sharp ramp seen at a single plant, so ramp results will look easier than they really are. Ramp events are also rare, so those results will have large error bars. I will report both of these openly. Code, settings, and fold dates will be shared to allow for reproducibility of the study.

Research Topic

Why does this matter now? Because solar keeps growing its share of electricity, and the costs of forecasting mistakes get higher as it does. These forecasts make or lose money for grid operators, plant owners, and energy traders. If skill is inflated as reported, they may end up trusting a model that does not help them.

This study provides three things: a fair and repeatable way to test solar forecasts, a measured number of how much random splitting inflates results, and a clear answer on the horizon where a model stops adding value over a simple baseline. They are useful even if the finding is that models help less than papers claim. The next logical step is a dashboard with sudden-drop alerts and explanations of each prediction, but it is outside the scope of this proposal.

Quality of Writing

I wrote this proposal in plain language and explained technical terms when they first appeared. References and in-text citations are in IEEE format, where sources are numbered as they are first cited.

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